

COURSE TITLE : POLYMER ANALYSIS LAB

COURSE CODE : 3098

COURSE CATEGORY : A

PERIODS/WEEK : 4

PERIODS/SEMESTER: 60/3

CREDITS : 3

TIME SCHEDULE

Module	Topics	Period
1	Identification of polymers	30
2	Purity of compounding ingredients	20
3	Iodine number of carbon black	10
TOTAL		60

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To identify raw rubbers and plastics by chemical analysis
2	1	To determine the purity of compounding ingredients by specific gravity,moisture content,molecular weight,viscosity,ash content and acid value.
3	1	To estimate Iodine number of carbon black

LIST OF EXPERIMENTS

MODULE I IDENTIFICATION OF POLYMERS

1.1.0 To identify raw rubbers and plastics by chemical analysis

1.1.1. Identification of raw rubbers-SBR,BR,EPDM,CR, NBR,IIR,HYPALON,THIOKOL and SILICONE

1.1.2. Identification of plastics-PP,PE,PVC and nylons

MODULE II PURITY OF COMPOUNDING INGREDIENTS

2.1.0 To determine the purity of compounding ingredients by specific gravity,moisture content,molecular weight,viscosity,ash content and acid value.

2.1.1. Determine the percentage purity of MBT

2.1.2. Determine the percentage purity of DPG

2.1.3. Determine the moisture content of clay

2.1.4. Determine the acid value of stearic acid.

MODULE III IODINE NUMBER OF CARBON BLACK

- 3.1.0 To estimate iodine number of carbon black
 - 3.1.1. Find out the iodine number of HAF black
 - 3.1.2. Find out the iodine number of GPF black